

32. (1) 记 X 为检修前生产的零件数, Y 为检修前共生产数

$$P(X=i|Y=n) = C_{n-1}^i \left(\frac{p_1}{p_1+p_2}\right)^i \left(\frac{p_2}{p_1+p_2}\right)^{n-1-i}$$

$$(i \leq n)$$

$$\text{即 } X|Y=n \sim B(n, \frac{p_1}{p_1+p_2}) \therefore E(X|Y=n) = \frac{p_1}{p_1+p_2} (n-1)$$

$$(2) E(X) = \sum_{i=1}^{\infty} E(X|Y=i) P(Y=i) = \frac{p_1}{p_2}$$

35.

$X \backslash Y$	2	3	4
0	$\frac{3}{7}$	0	0
1	$\frac{1}{7}$	0	0
2	0	$\frac{1}{7}$	$\frac{2}{7}$

$$\text{最佳均方预期即为 } E(L|X) = \begin{cases} 2 & X=0,1 \\ \frac{11}{3} & X=2 \end{cases}$$

$$36. \text{ 证 } E(Y - E(Y|X)) = 0$$

$$\text{证 } Z = Y - E(Y|X)$$

$$\text{由 } E((X - \varphi(Y))^2) = E((X - E(X|Y))^2) + E((E(X|Y) - \varphi(Y))^2)$$

$$\text{取 } \varphi(Y) = E(X)$$

$$\text{则 } E((X - E(X))^2) = DX$$

又

$$\begin{aligned} & E((E(X|Y) - E(X))^2) \\ &= E((E(X|Y) - E(E(X|Y)))^2) \\ &= D(E(X|Y)) \end{aligned}$$

$$\therefore DX = E(D(X|Y)) + D(E(X|Y))$$

$$37. (1) P(X_n \geq 0, n=1, 2, 3, 4) \\ = P(p + pq, p) = p^2(1+pq)$$

$$(4) P(|X_n| \leq 2, n=1, 2, 3, 4) \\ = 1 - P(\exists n=1, 2, 3, 4, |X_n| > 2) \\ = 1 - P(\exists n=3, 4, |X_n| > 2) \\ = 3pq$$

$$39. (1) P(X_{n+1} = i_{n+1} | X_n = i_n, \dots, X_0 = i_0) = \frac{P(X_{n+1} = i_{n+1}, \dots, X_0 = i_0)}{P(X_n = i_n, \dots, X_0 = i_0)} \\ = \frac{P(X_{n+1} - X_n = i_{n+1} - i_n) \dots P(X_1 - X_0 = i_1 - i_0) P(X_0 = i_0)}{P(X_n - X_{n-1} = i_n - i_{n-1}) \dots P(X_0 = i_0)} \\ = P(X_{n+1} - X_n = i_{n+1} - i_n) \\ = P(X_{n+1} = i_{n+1} | X_n = i_n)$$

$$\frac{(1) P(X_{n+1} = i_{n+1}, X_n = i_n, X_{n-1} = i_{n-1})}{P(X_n = i_n)} \\ = \frac{P(X_{n+1} - X_n = i_{n+1} - i_n) P(X_n = i_n) P(X_{n-1} - X_{n-2} = i_{n-1} - i_{n-2})}{P(X_n = i_n)}$$

$$= P(X_{n+1} - X_n = i_{n+1} - i_n) P(X_{n-1} - X_{n-2} = i_{n-1} - i_{n-2}) \\ = P(X_{n+1} = i_{n+1} | X_n = i_n) P(X_{n-1} = i_{n-1} | X_n = i_n)$$

已知:

$$1. P(X=2) = \frac{1}{3}$$

$$P(X=5) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

$$P(X=7) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

$$P(X=10) = \frac{1}{3}$$

$$\therefore E^*X = 6$$

2. 设 Y 为 6 点前出现的点数, 则 $X|Y=i \sim B(i, \frac{1}{5})$

$$\text{则 } E(X|Y=i) = \frac{i}{5}$$

$i=0, 1, \dots$

$$E^*X = \sum_{i=0}^{\infty} E(X|Y=i) P(Y=i) = 1$$

$$\frac{1}{1} + \frac{5}{6} + \frac{2}{6} + \frac{10}{7}$$